



Managerial Economics

A Problem-Solving Approach

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games and concluded that a rational reciprocity underlies people's motivation. This indicates that it pays to have morality and emotions, because the resulting 'irrational' behaviour is actually in our rational self-interest in the long run. The challenge is to be able to model these more complicated patterns of behaviour.

3.4 Factors determining demand

This section is concerned with identifying the factors which affect the quantity demanded of a product, describing the nature of these factors, examining how they can be measured, and examining the relationship with quantity demanded. These factors can be considered in terms of a demand function:

$$Q = f(P, L, A, D, Y, \dots) \quad (3.15)$$

where each symbol on the righthand side denotes a relevant factor. P refers to price, L to quality, A to advertising spending, D to distribution spending and Y to the average income of the market.

It is useful from a managerial decision-making viewpoint to distinguish between controllable and uncontrollable factors. Controllable in this context means controllable by the firm; the distinction in practice between what can be controlled and what cannot is somewhat blurred, as will be seen. The relevant factors are summarized in Figure 3.7.

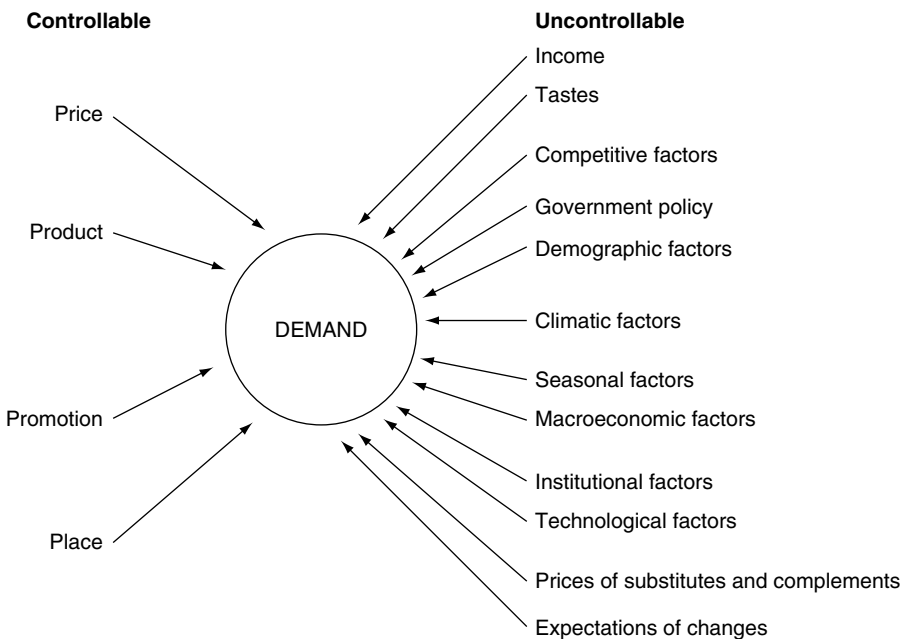


Figure 3.7. Factors determining demand.

3.4.1 Controllable factors

These correspond to what are often referred to as the marketing-mix variables.

a. Price

The nature of price has been discussed above, in terms of how it can be measured and how it affects the quantity demanded. A further aspect of the nature of price, as an indicator of quality, will be considered in Chapter 4, in terms of the implications of empirical studies. The firm may well be a multi-product firm, in which case the prices of its other products are also relevant, since they are often substitutes or complements. This factor is discussed in the context of 'uncontrollable factors'.

b. Product

It is the perceived quality of the product that affects quantity demanded. In practice there are two problems of measurement. The first of these is subjectivity: different consumers perceive quality differently. The second is multidimensionality which is now discussed at greater length.

Products often have many different attributes, as was mentioned in the previous section. This is particularly obvious for expensive and hi-tech products. With mobile phones, for example, a consumer may be looking for a good design, small size and weight, multiple features, a good complementary network, a good brand name, a good warranty, and so on. Even with an inexpensive product like toothpaste, consumers may be looking not just for something to clean their teeth, but also something that whitens their teeth, smells fresh, prevents decay and plaque and comes in a convenient dispenser. Marketing professionals and researchers often make use of multidimensional scaling in this situation. Consumers are asked to rate a product on a range of characteristics, giving a score for each one. Such questionnaires have to be carefully designed in order to include all the relevant factors; they also involve assumptions regarding the relative importance of the factors in terms of their weighting. Sometimes the unit cost of production is used as an approximate measure of quality; this again may not be an accurate indicator of quality, since it may be more a reflection of the efficiency of the firm.

c. Promotion

This refers to communication by the firm with its intended customers, ultimately aimed at persuading them to buy the product. The main components of the promotion-mix are customarily identified as: advertising, personal selling, publicity and sales promotion. Sometimes other elements are included separately, like direct marketing or packaging. These are usually measured in terms of expenditure and there is a direct relationship with quantity demanded.

d. Place

This is not a very apt description of what is involved here. It refers to distribution strategy, in particular the selection of distribution channels. This is again measured in terms of expenditure and again there is a direct relationship with quantity demanded.

In summary, the last three components of the marketing mix can all be considered as costs, while price is the only component that contributes directly to revenue; its relationship with revenue depends on price elasticity and this is examined in section 3.5.

3.4.2 Uncontrollable factors

These generally are external or environmental, and include a wide variety of factors whose relative importance varies from one product to another.

a. Income

This is a very important factor affecting quantity demanded for the majority of goods. Although it is expressed numerically there are a number of measurement problems. First of all, economists measure income in a number of different ways. The measure that is usually the most relevant to determining demand is personal disposable income (PDI). Although most governments or governmental agencies publish reliable figures relating to this on a regular basis, these are on a national basis and are therefore only strictly applicable to firms that have national markets. It is the average personal disposable income of the market for the firm that is relevant. There is thus a measurement problem if the firm has only a small market, because of the difficulty in obtaining relevant information; this is easier for national markets, or at least regional markets. In the situation where a firm has a small market, or one that is difficult to identify in terms of geographical area or otherwise, some proxy income variable may have to be used. For example, a shop selling products to just a small area or customer base within London may have to use average income in London as a proxy variable. The relationship with quantity demanded can be direct or inverse depending on whether the product is normal or inferior, as discussed in the previous section.

b. Tastes

These are difficult to define and measure, because they relate to subjective preferences and are multidimensional, just like perceived quality, which they influence. They can vary considerably both **intertemporally** and **inter-spatially**. **Intertemporal variations are variations over time**; these apply especially to products whose demand is affected by fashion. This refers not just to clothes, but to other consumer products like entertainment, food and sports. Examples of products where fashion has caused big increases in demand are

micro-scooters, teletubby dolls (Barney dolls in the United States), computer games, adventure sports and Viagra. Of course, demand for these products can also fall rapidly when they go out of fashion, like a yo-yo (another good example). These trends in fashion may be recurrent over time.

Inter-spatial variations are variations from place to place, especially from country to country. Good examples of products where tastes vary from country to country are beer, Coke, ketchup, TV programmes and Marks & Spencer (see Case 3.1). Much of this variation arises because of socio-cultural factors. Of course, tastes are not an entirely uncontrollable factor; firms try to influence tastes through their advertising. Cigarette advertising is an example of this, and this is one factor that causes governments to regulate it. This leads us straight into the next factor.

c. Government policy

This has both micro- and macroeconomic effects. The first of these are considered here. Governments often want to discourage us from buying certain products and sometimes to encourage us to buy more of other products. The reasons for this are examined in Chapter 12. Examples of the first category are cigarettes, alcohol, many other drugs (as opposed to medicines, which are in the second category, even though in practice the distinction may be very difficult to draw), weapons, products whose consumption or production causes environmental damage, and sometimes pornography and foreign goods. Goods in the second category are often their opposites. There are many policy instruments in either case: bans, indirect taxes/subsidies, health warnings, promotion regulations, licences, age restrictions, restrictions on times and places of consumption. It is important to distinguish between policies primarily affecting supply, like the first two, and those affecting demand. The fact that indirect taxes (or subsidies) do not affect demand is explained in the next subsection.

d. Competitive factors

This refers to the marketing mix of competitors. Price is not the only relevant factor here because firms often compete in other ways. This non-price competition occurs particularly in oligopolistic markets, for reasons that will be explained in Chapter 8. An example of this is the competition between Coke and Pepsi, who have a virtual duopoly of the cola market. The competition between them occurs mainly in terms of promotion and distribution.

e. Demographic factors

These refer not only to the size of population, but also to its structure. In particular the ageing of the population affects demand in many ways because older people demand more health services and pensions. This has important implications for government policy and the taxpayer.

f. Climatic factors

Weather, rainfall, temperature and also terrain are important. In Canada, for example, there is a greater demand for snow shovels, snow tyres and chains, salt for the roads and rust-proofing for cars.

g. Seasonal factors

These refer to any regular, repeated pattern in demand. Many products have a demand that varies according to season of the year, like air travel, hotels, car rental, jewellery and restaurants. The latter also has a monthly pattern (more people eat out at the beginning of the month when they have more cash), a weekly pattern (more people eat out at weekends) and a daily pattern. Many products have a daily pattern in demand, for example public transport, electricity, telephone calls and health clubs. This also has implications for pricing; many firms use peak and off-peak pricing. This will be examined in more detail in Chapter 10.

h. Macroeconomic factors

These include income, discussed earlier, and also interest rates, exchange rates, inflation rates and unemployment rates. If interest rates fall, for example, this may affect demand for two reasons: many homeowners have mortgages and the falling interest rate will increase their **discretionary income**. This is the income that they have available to buy non-necessities. Thus they will buy more of most (normal) goods. In addition, the demand for consumer durables will be increased, since these are often bought on credit and will therefore be in effect cheaper. Government policy obviously affects these macroeconomic factors to some degree.

i. Institutional factors

These include a wide variety of effects; the physical infrastructure of roads, railways and telecommunications is relevant, so are political systems, legal systems, education systems, housing systems, religious systems and family systems. For example, a country that has a poor transportation infrastructure will have a low demand for cars; a country with low literacy levels will have a relatively low demand for newspapers and magazines.

j. Technological factors

These primarily affect supply but also affect demand indirectly; this is best explained with an example. Mobile phones will again serve this purpose. Before the 1980s the concept of a truly mobile phone was still in the realm of science fiction, because the technology was not available. Thus consumers had not taken time to consider the potential of such a product and there was no real demand. As advancing technology made possible smaller, lighter, more effective and cheaper products, consumers had a true chance to evaluate the potential of the product and then demand began to increase accordingly.

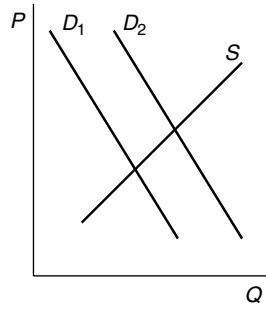


Figure 3.8. Change in demand.

k. Price of substitutes and complements

A substitute is a product with similar perceived functions or attributes; cars and public transport are substitutes. If the price of public transport were to rise this would increase the demand for cars, other things being equal. Complementary products are consumed together in some way; cars and petrol are complements. If the price of petrol were to rise this would reduce the demand for cars, other things being equal. Thus in the case of substitutes the relationship is positive, while in the case of complements the relationship is negative.

l. Expectations of changes in any of the above factors

If any factor, controllable or uncontrollable, is expected to change in the future, this can affect current sales. For example, it has been estimated that car sales in the UK in 2000 were 300,000 units lower than they otherwise would have been because prices were expected to fall. Similarly, expectations of changes in income, climate, government policy and competitive factors can all affect current demand.

3.4.3 Demand and quantity demanded

As already mentioned in the first section of this chapter, the failure to distinguish between these is very common and causes errors of analysis. The following explanation, graphs and example are intended to clarify this distinction.

a. Type of change: demand

- 1 *Graphical Representation*: Shifts in the demand curve (Figure 3.8)
- 2 *Cause*: Changes in any factor affecting the quantity demanded apart from price.

b. Type of change: quantity demanded

- 1 *Graphical representation*: Movements along the demand curve (Figure 3.9)
- 2 *Cause*: Changes in price, which are in turn caused by changes in costs and supply.

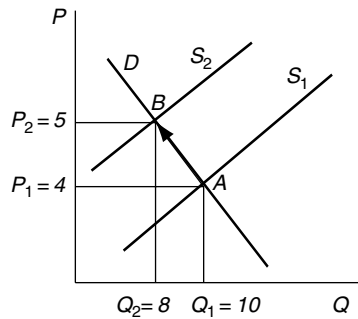


Figure 3.9. Change in quantity demanded.

A good example illustrating the difference between the two cases is the effect of an indirect tax. It is sometimes claimed that a tax on cigarettes will reduce the demand. This is untrue; such a claim would relate to the first case above (a change in demand), but with the demand curve shifting to the left, *reducing* the price. In fact we know that the price increases. This is because the increase in an indirect tax falls into the second category above (a change in quantity demanded). It increases the marginal costs of production, thus shifting the supply curve upwards and to the left. This increases the price (from P_1 to P_2) and reduces the quantity demanded (from Q_1 to Q_2).

Case study 3.1: Marks & Spencer

Does M&S have a future?¹³

The country's most famous retailer Marks & Spencer's big store in London's Kensington High Street has just had a re-fit. Instead of the usual drab M&S interior, it is now Californian shopping mall meets modernist chrome and creamy marble floors. Roomy walkways and designer displays have replaced dreary row after row of clothes racks. By the end of the year M&S will have 26 such stores around Britain – the first visible sign that the company is making a serious effort to pull out of the nose-dive it has been in for the past two years.

Things have become so bad that M&S, until recently a national icon, is in danger of becoming a national joke. It does not help that its advertisements featuring plump naked women on mountains – the first-ever TV ads the company has produced – have met with an embarrassed titter; nor that, last week, the BBC's Watchdog programme savaged M&S for overcharging and poor quality in its range of garments for the fuller figure.

As the attacks grow in intensity, so do the doubts about M&S's ability to protect its core value: a reputation for better quality that justified a slight price premium – at least in basic items, such as underwear. It is a long time since any self-respecting teenager went willingly into an M&S store to buy clothes. Now even parents have learned to say no. Shoppers in their thirties and forties used to dress like their parents. Now many of them want to dress like their kids.

M&S's makeover comes not a moment too soon. Compared with the jazzy store layouts of rivals such as Gap or Hennes & Mauritz, M&S shops look like a hangover from a bygone era. The makeover aims to bring it into the present.

People tended to join M&S straight from college and work their way slowly up the ranks. Few senior appointments were made from outside the company. This meant that the company rested on its laurels, harking back to 'innovations' such as machine-washable pullovers and chilled food.

Worse, M&S missed out on the retailing revolution that began in the mid-1980s, when the likes of Gap and Next shook up the industry with attractive displays and marketing gimmicks. Their supply chains were overhauled to provide what customers were actually buying – a surprisingly radical idea at the time.

M&S, by contrast, continued with an outdated business model. It clung to its 'Buy British' policy and it based its buying decisions too rigidly on its own buyers' guesses about what ranges of clothes would sell, rather than reacting quickly to results from the tills. Meanwhile, its competitors were putting together global purchasing networks that were not only more responsive, but were not locked into high costs linked to the strength of sterling.

In clothing, moreover, M&S faces problems that cannot be solved simply by improving its fashion judgments. Research indicates that overall demand for clothing has at best stabilised and may be set to decline. This is because changing demographics mean that an ever-higher share of consumer spending is being done by the affluent over-45s. They are less inclined than youngsters to spend a high proportion of their disposable income on clothes.

The results of M&S's rigid management approach were not confined to clothes. The company got an enormous boost 30 years ago when it spotted a gap in the food market, and started selling fancy convenience foods. Its success in this area capitalised on the fact that, compared with clothes, food generates high revenues per square metre of floor space. While food takes up 15% of the floor space in M&S's stores, it accounts for around 40% of sales. But the company gradually lost its advantage as mainstream food chains copied its formula. M&S's share of the British grocery market is under 3% and falling, compared with around 18% for its biggest supermarket rival, Tesco.

M&S has been unable to respond to this competitive challenge. In fact, rather than leading the

way, it has been copying rivals' features by introducing in-house bakeries, delicatessens and meat counters. Food sales have been sluggish, and operating margins have fallen as a result of the extra space and staff needed for these services. Operating profits from food fell from £247m in 1997 to £137m in 1999, while sales stayed flat.

Perhaps the most egregious example of the company's insularity was the way it held out for more than 20 years against the use of credit cards, launching its own store card instead. This was the cornerstone of a new financial-services division, also selling personal loans, insurance and unit-trust investments. When, in April this year, M&S eventually bowed to the inevitable and began accepting credit cards, it stumbled yet again. It had to give away around 3% of its revenues from card transactions to the card companies, but failed to generate a big enough increase in sales to offset this. Worse, it had to slash the interest rate on its own card, undermining the core of its own finance business. And this at a time when the credit-card business was already becoming more competitive, with new entrants offering rates as low as 5%.

If shrunk to its profitable core, M&S may become an attractive target for another big retailer. At the moment, however, while its food division may be attractive to the likes of Tesco, the clothing side represents a daunting challenge. Why take the risk now, when the brand may be damaged beyond repair?

Questions

- 1 Identify the main factors affecting the demand for M&S products.
- 2 Analyse the weaknesses and threats on the demand side of M&S, relating these to controllable and uncontrollable factors.

3.5 Elasticity

Elasticity in general terms is concerned with **the responsiveness or sensitivity of one variable to changes in another**. Demand elasticity is concerned with how the quantity demanded is responsive to changes in any of the factors described in section 3.4. Essentially elasticities are an alternative way of measuring responsiveness to the marginal effects discussed in the first section of this

chapter; however, they have the advantage that they allow more meaningful comparisons between different products because they are measured in relative terms, meaning percentages, rather than in absolute units. Theoretically we could discuss elasticities for all the different controllable and uncontrollable factors affecting demand but in practice there are four main types of elasticity that tend to be measured and examined, corresponding to four particularly important factors in the demand function: **price**, **promotion**, **income** and the **price of a related product**:

$$Q = f(P, A, Y, P') \quad (3.16)$$

Each of the above elasticities needs to be analysed in terms of:

- a. definition
- b. measurement
- c. range of values
- d. determinants
- e. use and applications.

3.5.1 Price elasticity

a. Definition

PED is the **percentage change in quantity demanded in response to a 1 per cent change in price**. In symbols we can write:

$$PED = \% \Delta Q / \% \Delta P \quad (3.17)$$

b. Measurement

The reader should be warned at this point that the following material will appear highly abstract at first, and for many people will only start to make sense when it is applied to the kinds of problem discussed later on in the chapter.

The definition in (3.16) is the easiest one to use for measurement. Referring to Figure 3.9, in the previous section, if price is raised from 4 to 5 units (the units do not matter):

$$PED \text{ from A to B} = -20 / +25 = -0.8.$$

The interpretation of this value is that for every 1 per cent rise in price there is a 0.8 per cent fall in the quantity demanded. However, there is a measurement problem here, because if we now reverse the price change and reduce the price to 4, the calculation is as follows:

$$PED \text{ from B to A} = +25 / -20 = -1.25.$$

Therefore this measure of PED (termed the **simple** PED) is **inconsistent**; later on it will also be shown to be **inaccurate**. The reason for this is the **changing**

base for the percentage calculation. It is therefore better in terms of consistency and accuracy to use a measure involving average prices and quantities. Thus

$$\% \Delta Q = \frac{\Delta Q}{(Q_1 + Q_2)/2} = -2/9 = -22.22$$

and

$$\% \Delta P = \frac{\Delta P}{(P_1 + P_2)/2} = 1/4.5 = 22.22$$

Therefore in the above example $PED = -1$, both from A to B and from B to A; this measure of PED is termed the **adjusted** PED. The procedure above can be simplified by deriving the formula:

$$\begin{aligned} PED &= \frac{\Delta Q}{\Delta P} \times \frac{P_1 + P_2}{Q_1 + Q_2} \\ &= \frac{-2}{1} \times \frac{4 + 5}{10 + 8} = -1 \text{ (as before)} \end{aligned} \quad (3.18)$$

It can easily be verified that the same result is obtained if the price change is from 5 to 4 units.

The above measures, both simple and adjusted, are called measures of **arc elasticity**, because they **apply between two points or values**. Sometimes we want to measure elasticity at a single point or price, and this is appropriate when we are considering the effect of a very small price change. This can be done using a **point elasticity**, and involves the concept of limits and derivatives. The relevant formula becomes:

$$PED = \frac{\partial Q}{\partial P} \times \frac{P}{Q} \quad (3.19)$$

where $\partial Q / \partial P$ represents the **partial derivative of Q with respect to P**. This mathematical concept of a partial derivative has an important economic interpretation, which follows from the material in section 3.2: it represents **the marginal effect of price on quantity, assuming other factors affecting demand remain the same**. Normally this partial derivative is calculated by differentiating the demand function, in this case:

$$Q = 18 - 2P$$

Therefore at A:

$$PED = -2 \times 4/10 = -0.8$$

and at B:

$$PED = -2 \times 5/8 = -1.25$$

This shows that the PED varies along the linear demand curve.

Table 3.3. Range of values for PED

$ PED $	Demand	Interpretation
>1	Elastic	Consumers responsive to price changes
<1	Inelastic	Consumers not very responsive to price changes
$=1$	Unit elastic	Intermediate case
∞	Perfectly elastic	Infinitely responsive (buy nothing if price rises)
0	Perfectly inelastic	Totally unresponsive (buy the same if price rises)

With the power form of demand curve we can again use either arc or point elasticity but in this case we can prove that they are the same. If we apply the point PED formula to the power form $Q = aP^b$ we obtain the following from (3.19), using partial differentiation:

$$PED = baP^{b-1} \times \frac{P}{Q} = baP^{b-1} \times \frac{P}{aP^b} = \frac{baP^b}{aP^b} = b \quad (3.20)$$

This is a very important result: it shows that for any power form of the demand curve the power represents the PED, which will be constant for all prices along the demand curve. This in turn means that the arc and point elasticity must be the same, as stated earlier. Furthermore, the result can be generalized to apply to any elasticity; **for any power form the power of a variable represents the elasticity of the dependent variable with respect to that independent variable.** This concept will be applied in many situations as we continue our analysis, not only in this chapter.

c. Range of values

This can be seen from Table 3.3. It should be noted that the PED is always interpreted in terms of its absolute magnitude when we are referring to whether demand is elastic or inelastic; its negativity is simply because of the inverse relationship between price and quantity demanded. We shall see, however, that in performing calculations involving the PED, it is important to include the sign.

d. Determinants of PED

There are three main factors that affect the price elasticity, of which the first is the most important.

1. Availability of substitutes. The more close a substitute(s) a product has the more elastic is its demand. An example is the demand for cigarettes: this demand is inelastic because such substitutes that exist (like cigars, pipes, chewing tobacco, other drugs, nicotine patches and chewing gum) are not very close in terms of their perceived functions and attributes. However, a single brand of cigarettes may have elastic demand, depending on brand loyalty. In this case we are referring to a different issue; instead of asking what will happen when the price of *all* cigarettes changes we are asking what

Table 3.4. Estimates of price elasticities

Product	PED
Airline tickets (USA–Europe), economy	–1.30 ¹⁴
Auto repair	–0.36 ¹⁵
Beef	–0.65 ¹⁶
Beer (consumed at home)	–0.84 ¹⁷
Bread	–0.09 ¹⁸
Cheese	–1.16 ¹⁸
Chicken	–0.65 ¹⁸
Chicken (not free range)	–0.13 ¹⁸
Cinema tickets	–0.87 ¹⁵
Coffee (instant)	–0.36 ¹⁹
Coffee (regular)	–0.16 ¹⁹
Dental visits (adults)	–0.72 ²⁰
Electricity (household)	–0.13 ¹⁵
Fruit juices	–0.80 ¹⁸
Furniture	–3.04 ²¹
Health club memberships	–0.87 ²²
Household appliances	–0.64 ¹⁵
Housing	–0.23 ²³
Legal services	–0.61 ¹⁵
Medical insurance	–0.31 ¹⁵
Milk	–0.18 ¹⁸
Milk	–0.49 ¹⁶
Potatoes	–0.27 ¹⁶
Restaurant meals	–1.63 ¹⁵
Shoes	–0.73 ¹⁵
Taxi service	–1.24 ¹⁵
Telephones	–0.10 ²⁴
Tobacco products	–0.46 ¹⁵

will happen when the price of one brand changes *while the price of other brands stays the same*. In this situation the PED will depend on whether other brands are seen as substitutes; hence the importance of brand loyalty in affecting PED. If there is a great amount of brand loyalty, other brands will not be seen as substitutes and demand will be less elastic than would be the case if there were less brand loyalty. Generally speaking, the more narrowly a product is defined the more elastic its demand will be, for example food, breakfast cereals, cornflakes and Kellogg's cornflakes have increasingly elastic demand.

2. Proportion of income spent on the product. The higher this is, the more elastic the demand, other things remaining equal. This is because the income effect of the price change is greater. Thus the demand for sugar will tend to be inelastic while the demand for cars will be more elastic.

3. Time frame. Demand tends to be more elastic in the longer term, because it may take time for consumers to switch to different products. An example of this was oil in the 1970s, when the price quadrupled. The demand was inelastic in the short run because it was difficult for households and firms to cut back on their consumption. As time went on consumers could switch to other forms of heating and more fuel-efficient cars, and firms also switched to other firms of energy, such as electricity and gas-turbine generators. Habits are easier to change over a longer time period. Two further examples relate to electricity and cinema tickets. The

Table 3.5. PED relationships with revenue, costs and profit

Demand	Price	Quantity	Revenue	Costs	Profit
Elastic	↑	↓ more	↓	↓	?
	↓	↑ more	↑	↑	?
Inelastic	↑	↓ less	↑	↓	↑
	↓	↑ less	↓	↑	↓
Unit elastic	↑	↓ same	same	↓	↑
	↓	↑ same	same	↑	↓

↑ = increases; ↑ more = increases by a larger proportion; ↓ less = increases by a smaller proportion; ↑ same = increases by the same proportion; ↓ = decreases;

↓ more = decreases by a larger proportion; ↓ less = decreases by a smaller proportion;

↓ same = decreases by the same proportion; ? = unknown without further information

PEDs of -0.13 and -0.87 respectively, as reported in Table 3.4, are short-run estimates; the same study estimated the long-run PEDs at -1.89 and -3.67 .

These examples of short-run and long-run estimates illustrate a very important point, one that is emphasized throughout this text: theory alone cannot predict exactly what the PED for a product will be; it can only provide some indications of the general size and sign of the value. We need to measure elasticities using empirical studies (this methodology will be explained in Chapter 4) to confirm (or refute) any theoretical considerations. Table 3.4 shows selected results for a wide variety of different products from a number of different studies in different countries. Two products, chicken and milk, are included in the table twice because different estimates have been obtained in different studies.

e. Use and applications

Firms want to know the PED for their products in order to charge the right price and to make forecasts. The 'right' price is the price that achieves the firm's objectives, for example profit maximization or revenue maximization. Some important relationships are shown in Table 3.5, which shows the effects of price changes on other variables, given different values of the PED. At this stage the topic of costs has not been covered, but because the table is very general in terms of examining the directions of the changes rather than their precise sizes, it is sufficient to note that costs (meaning total costs) are directly related to output. It is assumed here that this output corresponds to the quantity that people want to buy, implying that firms maintain fairly constant levels of inventory.

The above information can also be illustrated graphically. Figure 3.10 aids in understanding the relationships between the variables and the implications involved. Note that a linear demand curve is assumed, with an equation corresponding to the graph in Figure 3.9; this is given by $Q = 18 - 2P$, which readers should verify for themselves. Table 3.6 shows the relationships in Figure 3.9 in numerical form, ignoring costs. It should be noted that the marginal revenue can be calculated either from taking the differences in total revenue between

Table 3.6. Linear demand, PED and revenue

<i>Q</i>	<i>P</i>	<i>TR</i>	<i>MR</i>	<i>PED</i>
0	9	0	0	$-\infty$
2	8	16	7	-8
4	7	28	5	-3.5
6	6	36	3	-2
8	5	40	1	-1.25
10	4	40	-1	-0.8
12	3	36	-3	-0.52
14	2	28	-5	-0.29
16	1	16	-7	-0.125
18	0	0	-9	0

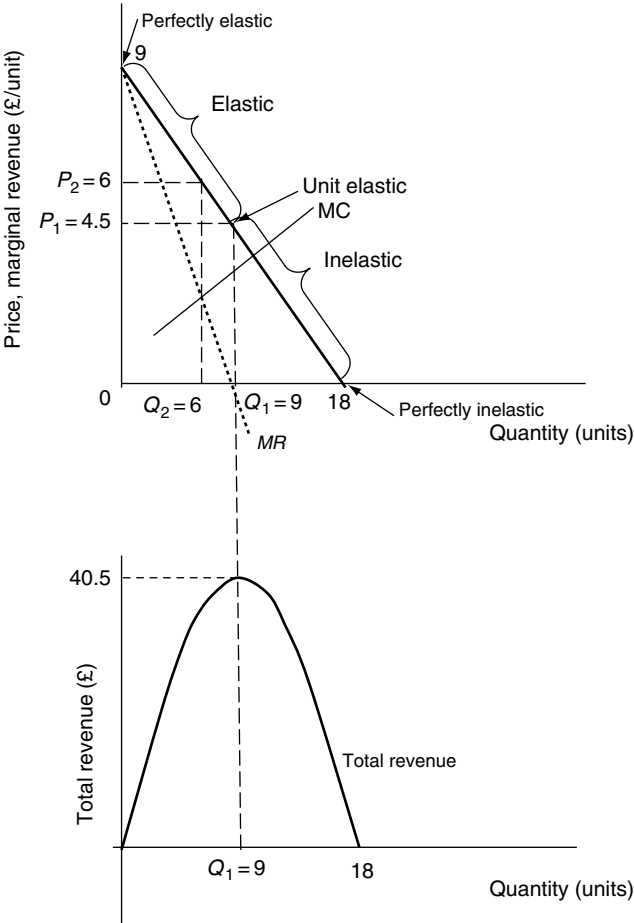


Figure 3.10. PED, revenue and profit.

each pair of prices or by taking the derivative of the revenue function; the latter method is used in Table 3.6.

Tables 3.5 and 3.6 and Figure 3.10 show three things of particular importance:

1. The arc elasticity using average prices and quantities is correct. If revenue stays the same when there is a price change, demand must have unit elasticity; revenue at A and B in the earlier example was the same, at £40. This can be shown more formally by having a power form $Q = aP^{-1}$; this can be rewritten $Q = a/P$, where a is a constant representing revenue.

2. Firms will always maximize revenue by charging a price (P_1) where demand is unit elastic. This is proved mathematically in Chapter 8. In this situation $MR = 0$. It can also be seen that the MR curve has twice the slope of the demand curve. Thus the output Q_1 is halfway between the origin and point B, the horizontal intercept of the demand curve. This is proved as follows:

The demand curve has equation $P = a + bQ$

Revenue, $R = PQ = (a + bQ)Q = aQ + bQ^2$

The MR curve has equation $MR = \frac{dR}{dQ} = a + 2bQ$ (3.21)

The interpretation of the MR curve and its shape is considered in more detail in Chapter 8.

3. Firms will always maximize profit by charging a price (P_2) where demand is elastic. This can be explained in terms of Figure 3.10 by seeing that $MC = MR$ (the condition for profit-maximizing output) at an output less than Q_1 , since MR must be positive in this situation. Therefore at this output (Q_2) the price must be higher than P_1 , so demand must be in the elastic range. In order to determine the exact price that maximizes profit, more precise information regarding the demand and cost functions must be known; this is examined in Chapter 8.

As far as forecasts are concerned, the firm can estimate the effects on sales of a change in price if it knows the PED. These effects are shown in detail in the worked problems.

In the above analysis we concentrated on the importance of the PED from the viewpoint of the management of the firm. Governments are also interested in price elasticities as a guide to various aspects of policy, for example the determination of levels of indirect taxes and import duties, and competition policy. These aspects are considered in Chapter 12.

3.5.2 Promotional elasticity

Promotional elasticity is now discussed; this is not because it is seen as being next in importance to price elasticity, since that position usually is taken by

income elasticity, but because, like price, it is a **controllable variable**. Since, in practice, promotion often refers to advertising, the term ‘advertising elasticity’ will be used from now on.

a. Definition

Advertising elasticity (AED) can be defined in a very similar way to PED; AED refers to the **percentage change in quantity demanded in response to a 1 per cent change in advertising**. This means just changing the last word of the definition of PED to advertising (measured as expenditure). In symbols:

$$AED = \% \Delta Q / \% \Delta A \quad (3.22)$$

b. Measurement

As with PED both arc and point elasticities can be used. The **simple** measure of elasticity, using the formula above, should again not normally be used, unless the changes in advertising are very small, because it is **inconsistent** and **inaccurate**. The formulae for arc and point elasticities are similar to the formulae for PED, with the variable A substituted for P . Thus we have the following:

$$\text{Adjusted AED} = \frac{\Delta Q}{\Delta A} \times \frac{A_1 + A_2}{Q_1 + Q_2} \quad (3.23)$$

$$\text{Point AED} = \frac{\partial Q}{\partial A} \times \frac{A}{Q} \quad (3.24)$$

where $\partial Q / \partial A$ represents the partial derivative of Q with respect to A . This partial derivative has a similar economic interpretation to that involved in price elasticity; in this case it represents the marginal effect of advertising on quantity, assuming that other factors remain constant. With the power form of demand equation $Q = aP^b A^c$, it can be shown using partial differentiation that the advertising elasticity is given by c . We can see that all elasticities in a power form are given by the powers of the relevant variables and are constant; thus the power form is sometimes known as the **constant elasticity** form.

c. Range of values

AED is expected to be positive, since we would expect consumers to react positively to promotion or advertising.

d. Determinants

The **effectiveness** of the firm’s advertising is important and so is the **amount spent** on advertising. As the firm spends more and more the market may become saturated (diminishing returns) and AED may decrease. There are no widely published results of empirical studies relating to promotional elasticities, and it would be difficult to generalize any conclusions even if there were.

Table 3.7. Range of values for YED

YED	Demand	Interpretation
> 1	Income elastic	Luxury products
$0 < YED < 1$	Income inelastic	Staple products
< 0	Negative elasticity	Inferior products

e. Use and applications

Firms want to know the AED for their products in order to spend the right amount on promotion and to make forecasts. The details of how to optimize promotion are given in Chapter 10, in the section on pricing in the marketing mix.

3.5.3 Income elasticity

a. Definition

Income elasticity (YED) is defined similarly to the above elasticities; it is the **percentage change in quantity demanded in response to a 1 per cent change in income**. Again just the last word is changed, to 'income'. In symbols:

$$YED = \% \Delta Q / \% \Delta Y \quad (3.25)$$

b. Measurement

This is also the same as for the above elasticities; there are three measures, simple, adjusted and point, but the first should not normally be used unless the changes in income are very small. Thus the adjusted and point elasticities are:

$$\text{Adjusted YED} = \frac{\Delta Q}{\Delta Y} \times \frac{Y_1 + Y_2}{Q_1 + Q_2} \quad (3.26)$$

$$\text{Point YED} = \frac{\partial Q}{\partial Y} \times \frac{Y}{Q} \quad (3.27)$$

where $\partial Q / \partial Y$ represents the partial derivative of Q with respect to Y . The power form of the demand equation would be extended to become $Q = aP^b A^c Y^d$.

c. Range of values

Income elasticity can be positive or negative, depending on whether the product is normal or inferior. Normal products can be further divided into **luxury** products and **staple** products, according to whether the income elasticity is **more than one** or **less than one** respectively. This is summarized in Table 3.7.

d. Determinants

As can be seen above, the **type of product** is the main factor determining the income elasticity. The **level of income** in the market is also relevant; for

Table 3.8. Estimates of income Elasticities

Product	YED
Airline tickets (USA–Europe), economy	1.38 ¹⁴
Alcohol	1.54 ¹⁵
Apples	1.32 ¹⁶
Beef	1.05 ¹⁶
Bread	−0.25 ²⁵
Butter	0.42 ¹⁵
Car travel	1.23 ²³
Chicken	0.28 ¹⁶
Clothing	1.47 ²³
Cinema tickets	3.41 ¹⁵
Coffee	0.00 ¹⁵
Dental visits (adults)	0.58 ²⁰
Flour	−0.36 ¹⁵
Furniture	1.48 ²¹
Housing (low-income renters)	0.22 ²⁶
Housing (owner-occupied)	1.49 ¹⁵
Margarine	−0.20 ¹⁵
Medical insurance	0.92 ¹⁵
Milk	0.50 ¹⁶
Potatoes	0.15 ¹⁶
Restaurant meals	1.40 ¹⁵
Shoes	1.10 ¹⁵
Tea	−0.56 ²⁵

example, products that are staple goods in developed countries, like telephones and radios, might be luxuries in a developing country. As with price elasticity, theory alone cannot tell us whether a product is in a certain category; we need empirical studies to confirm this. Table 3.8 shows selected results for a wide variety of different products from a number of different studies in different countries.

e. Use and applications

Firms want to know the income elasticities for their products in order to select target markets and make forecasts. Note that income is not truly a controllable variable, but a firm can gain an element of control by selecting from different target markets with different levels of average income.

3.5.4 Cross-elasticity

a. Definition

This refers to the **percentage change in the quantity demanded of one product in response to a 1 per cent change in the price of another product**. This can be expressed in symbols:

$$CED = \% \Delta Q / \% \Delta P' \quad (3.28)$$

b. Measurement

The same methods and principles apply as for the other elasticities. Thus:

$$\text{Adjusted CED} = \frac{\Delta Q}{\Delta P'} \times \frac{P'_1 + P'_2}{Q_1 + Q_2} \quad (3.29)$$

$$\text{Point CED} = \frac{\partial Q}{\partial P'} \times \frac{P'}{Q} \quad (3.30)$$

where $\partial Q / \partial P'$ represents the partial derivative of Q with respect to P' .

The power form of the demand equation would be extended to become $Q = aP^b A^c Y^d P'^e$.

c. Range of values

CED can be **positive** or **negative** according to whether the other product is a **substitute** or a **complement**. For example, if the price of Pepsi rises, the quantity of Coke demanded should also rise, as people switch from Pepsi to Coke. Note the importance of the assumption of other things being equal, in particular that the price of Coke stays the same.

d. Determinants

As well as the above, the **degree of substitutability or complementarity** is important. The greater this is, the larger the absolute value of CED. Thus, in the example above, the closer Pepsi and Coke are seen as substitutes, the more people will switch from one to the other. Again, it is important to test theoretical considerations by empirical studies. For example, the CED between electricity and natural gas has been estimated to be only 0.20 (US data);²⁷ the same study estimated the CED between butter and margarine to be 0.67 (when the price of margarine changed), while another study²⁸ estimated this CED to be 1.53 (when the price of butter changed). The last two products are much closer as substitutes than gas and electricity.

Studies have also measured the CED for complementary products. For example, food has been found to be complementary to both clothing²⁹ and entertainment.³⁰ In the first case the CED was found to be -0.18 , while in the second it was -0.72 , indicating that food and entertainment are more complementary. In this case it does not mean that food and entertainment are actually consumed together (although this may be true); because food involves a relatively high proportion of income there is a substantial income effect when the price of food changes. When food prices rise, real incomes fall and this in turn causes a fall in spending on entertainment.

e. Use and applications

Firms want to know the CED for their products in order to see what their main competition is and determine strategy accordingly; sometimes the 'other products' involved are in the same product line or product mix of the firm itself. In this case, information relating to CEDs is important for product planning

purposes. The information is again useful for making forecasts, when the price of a related product changes, as shown in the solved problem [SP3.3](#).

3.6 A problem-solving approach

There is a world of difference between ‘knowing’ the principles of demand theory and being able to apply them to practical managerial problems, even of a simplified kind. Even though all problems in this area are essentially concerned with how different variables affect quantity demanded, this gulf between theory and application is greater with this topic than with most others since problems can come in many different forms and guises. Students are often bewildered by the problems and do not know how to start, even though they claim to know all the principles. The following approach is therefore recommended as an aid.

Three questions need to be asked, in the following order:

- 1 *Does the problem involve marginal effects or elasticities?* Sometimes this is explicit in the question, but in other cases the different interpretations of these different concepts, in terms of **units** or **percentages**, have to be applied to the data in the problem to see which is relevant.
- 2 *Which type of marginal effect or elasticity is relevant?* Again this may be specified in the question, but in other cases it is not, so the student has to decide which of the four variables is relevant: **price**, **promotion**, **income** or a **price of related good**. It is possible that a different variable may be involved, or that more than one variable is involved, for example in the solved problems [SP3.2](#) and [SP3.3](#).
- 3 *Which measure of elasticity is involved?* This obviously only relates to elasticity questions, since there is only one way of measuring marginal effects; however, elasticity can be measured using either **arc** or **point** methods. The main criterion is to see whether there is a **change** in the relevant variable, or whether it stays **constant**. In the first case the arc method is used, while in the second the point method is appropriate. In some questions either method is feasible (see [SP3.2](#) and [SP3.3](#)), but there are advantages and disadvantages of each, as will be explained.

Once the above questions have been answered it is recommended that the student write down the **relevant formula in general terms** rather than trying to substitute the numbers in the data immediately. This tends to eliminate a number of mistakes.

This approach is now illustrated in the following four solved problems.

SP3.1 Interpretation of elasticities

Midwest Cable TV has estimated the demand for its service to be given by the following function:

$$Q = 9.83P^{-1.2}A^{2.5}Y^{1.6}P_0^{-1.4}$$

where

Q = monthly sales in units

P = price of the service in \$

A = promotional expenditure in \$'000

Y = average income of the market in \$'000

P_0 = price of 'home movies' in \$

The current price of Midwest is \$60, promotional expenditure is \$120,000, average income is \$28,000, and the price of 'home movies' is \$45.

Indicate whether the following statements are true or false, giving your reasons and making the necessary corrections.

- If Midwest increases its price this will reduce the number of its customers.
- If Midwest increases its price this will reduce its revenues.
- People's expenditure on the cable TV service as a proportion of their income will increase when their income increases.
- If Midwest increases its price this will increase the sales of 'home movies'.
- 'Home movies' are a substitute for cable TV.
- A 5 per cent increase in income will increase demand by 16 per cent.
- A 10 per cent increase in price will reduce demand by 12 per cent.
- Current sales are over a million units a month.
- The demand curve for Midwest is given by: $Q = 9.83P^{-1.2}$
- Midwest's sales are more affected by the price of 'home movies' than by the price of its own service.
- If Midwest increases its price this will reduce its profit.

Solution

- True;** customers and quantity demanded are synonymous in this case, and there is an inverse relationship between Q and P , as seen by the **negative price elasticity**.
- True; demand is elastic**, since the PED is greater than 1 in absolute magnitude. Therefore an increase in price causes a greater than proportional decrease in quantity demanded and a fall in revenue.
- True;** this is because the YED is greater than 1, indicating that cable TV is a luxury product. Note that the statement would be false if the good were a staple. For staples, although expenditure on the product increases as income increases, expenditure as a proportion of income falls, since expenditure rises more slowly than income.
- False;** the two products are complementary, shown by the CED **being negative**; therefore an increase in the price of one product will reduce the sales of the other. It appears therefore that 'home movies' is a cable channel.
- False;** the two products are complementary, shown by the CED **being negative**.

- f. **False**; $YED = 1.6$; therefore using the simple elasticity formula (reasonably accurate for small changes) the **change in demand will be $1.6 \times 5\% = 8\%$** .
- g. **False**; change is $1.2 \times 10\% = 12\%$, but this is a **change in quantity demanded**, corresponding to a movement along a demand curve (unlike the previous part of the question, which involves a shift in the demand curve).
- h. **False**; current sales are given by
 $Q = 9.83(60)^{-1.2} (120)^{2.5} (28)^{1.6} (45)^{-1.4} = \mathbf{11,420 \text{ units}}$.
- i. **False**; the demand curve is given by
 $Q = 9.83P^{-1.2} (120)^{2.5} (28)^{1.6} (45)^{-1.4}$ or $Q = \mathbf{1,554,039P^{-1.2}}$.
- j. **True**; the **CED is larger in absolute magnitude than the PED**. This is an unusual situation but arises because of the nature of cable TV service. The service is only a means to an end, that of receiving certain channels.
- k. **False**; since **demand is elastic, an increase in price has an unknown effect on profit**. More information would be required.

SP3.2 Marginal effects and elasticity

PK Corp estimates that its demand function is as follows:

$$Q = 150 - 5.4P + 0.8A + 2.8Y - 1.2P^*$$

where

Q = quantity demanded per month (in 1000s)

P = price of the product (in £)

A = firm's advertising expenditure (in £'000 per month)

Y = per capita disposable income (in £'000)

P^* = price of BJ Corp (in £).

- During the next five years, per capita disposable income is expected to increase by £2,500. What effect will this have on the firm's sales?
- If PK wants to raise its price by enough to offset the effect of the increase in income, by how much must it raise its price?
- If PK raises its price by this amount, will it increase or decrease the PED? Explain.
- What is the relationship between PK and BJ? Explain your answer.
- If next year PK intends to charge £15 and spend £10,000 per month on promotion, while it believes per capita income will be £12,000 and BJ's price will be £3, calculate the income elasticity of demand. What does this tell you about the nature of PK's product?
- What effect would an increase in advertising of £1000 have on profitability, if each additional unit costs £10 to produce?

Solution

- a. Use the marginal effect of income on quantity demanded:

$$\Delta Q / \Delta Y = 2.8$$

$$\Delta Y = 2.5$$

$$\Delta Q / 2.5 = 2.8$$

$$\Delta Q = 2.8 \times 2.5 = 7, \text{ or } \mathbf{7,000 \text{ units}}$$

- b. Use the marginal effect of price on quantity demanded:

$$\Delta Q / \Delta P = -5.4$$

$$-7 / \Delta P = -5.4$$

$$\Delta P = -7 / -5.4 = \mathbf{\pounds 1.30}$$

- c. $PED = \partial Q / \partial P \times P / Q = -5.4P / Q$

As P increases and Q stays constant (because of rising income), PED increases in absolute magnitude, meaning that demand becomes **more elastic**.

- d. The two companies make **complementary products** because the marginal effect of the price of BJ on the quantity of PK is negative.

- e. $YED = \partial Q / \partial Y \times Y / Q$

$$= 2.8 \times 12 / (150 - 5.4 \times 15 + 0.8 \times 10 + 2.8 \times 12 - 1.2 \times 3)$$

$$= 2.8 \times 12 / 107 = \mathbf{0.3140}$$

(staple product)

- f. If $\Delta A = 1$, $\Delta Q = 0.8 = 800$ units, $\Delta R = 800 \times 15 = \pounds 12,000$

$$\Delta C = 800 \times 10 \text{ (production)} + \Delta A \text{ (\pounds 1,000)} = \pounds 9,000$$

Thus every additional \pounds 1,000 spent on advertising increases profit by \pounds 3,000.

This problem tests the ability of the student to distinguish between the concepts of marginal effect and elasticity, and apply each one accordingly. In part (f), in particular, it is important to use marginal effect, not elasticity, since profit is measured in money units. It is also important that the student observes the units in which the variables are measured.

SP3.3 Arc elasticities

LAD Enterprises sells mobile phones, currently charging \pounds 80 and earning \pounds 9,600 per week. It is considering a price cut of 20 per cent to increase its market share, estimating its PED to be -1.6 .

- a. Estimate the effect of the price cut on LAD's revenue, stating any assumptions.

- b. If a competitor responds to the price cut by also reducing its price by 20 per cent, estimate the effect on LAD's revenue, assuming the CED is 0.8.
- c. Compare the profits in (a) and (b) above with the original level of profit.

Solution

$$\text{a. } Q_1 = 9,600/80 = 120 \quad P_1 = 80 \quad P_2 = 64$$

$$-1.6 = \frac{Q_2 - 120}{-16} \times \frac{(80 + 64)}{120 + Q_2}$$

$$Q_2 = 172 \quad R_2 = 172 \times 64 = \mathbf{11,008}$$

$$\text{b. } 0.8 = \frac{Q_2 - 172}{-0.2P'} \times \frac{P' + 0.8P'}{172 + Q_2}$$

$$Q_2 = 144 \quad R_2 = 144 \times 64 = \mathbf{9,216}$$

- c. in (a), R rises, C rises, therefore effect on profit uncertain.
in (b), R falls, C rises, therefore profit falls.

This problem illustrates several points:

- 1 Arc elasticities can be used in a chained sequence where the Q_2 from one calculation becomes the Q_1 for the next calculation.
- 2 The effect of a competitor's price change can be calculated without knowing the level of price, as long as the percentage change is known.
- 3 A price war can often result in lower profit.

SP3.4 Point elasticities and power form

Pritti Sprays is a manufacturer of cosmetic products. Its management is currently engaged in an analysis of the lipsticks produced by the firm, examining the future demand for them. Market research indicates that advertising expenditure, price and average income are three major variables affecting the demand for the lipsticks. The advertising elasticity for demand is estimated to be 1.5, the price elasticity -1.2 and the income elasticity 1.8. In addition, the following data are available.

Year	Sales (units)	Advertising expenditure	Price	Income
2001	2,369	£21,000	£6.50	£25,000
2002		£21,000	£6.50	£27,000
2003		£24,000	£6.80	£29,000

- a. Estimate sales for 2002 and 2003.
- b. A competitor plans on reducing its price from £6.90 to £6.40 in 2003; assuming the CED is 1.5, how much would Pritti have to spend on advertising to achieve the same rate of growth of sales as from 2001 to 2002?

Solution

Since the elasticities are given as constants we can use these to calculate the power demand function:

$$Q = aP^{-1.2}A^{1.5}Y^{1.8}$$

$2369 = a(6.5)^{-1.2}(21^{(1.5)}25^{(1.8)})$ where promotion and income are measured in £'000

$$a = 0.7087$$

a. Sales in 2002 = $0.7087(6.5)^{-1.2}(21)^{1.5}(27)^{1.8} = \mathbf{2,721 \text{ units}}$

Sales in 2003 = $0.7087(6.8)^{-1.2}(24)^{1.5}(29)^{1.8} = \mathbf{3,581 \text{ units}}$

b. Growth from 2001 to 2002 = $(2,721/2,369) - 1 = 14.86\%$

Required sales in 2003 = 3,125

$$2,721 = a(6.5)^{-1.2}(21)^{1.5}(27)^{1.8}(6.9)^{1.5}$$

$$a = 0.0391$$

$$3,125 = 0.0391(6.8)^{-1.2}A^{1.5}(29)^{1.8}(6.4)^{1.5}$$

$$A^{1.5} = 114.8$$

$$A = 23.627; \quad \text{advertising} = \mathbf{£23,627}$$

This problem illustrates how the use of point elasticities in the power form can be used to approach a similar problem to SP3.2 when different independent variables change, assuming that the elasticities are constant. The advantage of the above method is that any number of variables can be changed simultaneously, whereas with the arc method only one variable can be changed at a time, thus lengthening the calculations.

Case study 3.2: The Oresund bridge**A not-so-popular Nordic bridge³¹**

It was not quite what the planners had in mind when Sweden and Denmark opened their expensive bridge across the Oresund strait in July. After an early boost from summer tourism, car crossings have fallen sharply, while trains now connecting Copenhagen, the Danish capital, and Malmo, Sweden's third city, are struggling to run on time. Many people think the costs of using the bridge are simply too high. And, from the point of view of Scandinavian solidarity, the traffic is embarrassingly one-sided: far more Swedes are going to Denmark than vice versa. So last week the authorities decided to knock almost 50% off the price of a one-way crossing for the last three months of this year.

The two governments, which paid nearly \$2 billion for the 16km (10-mile) state-owned bridge-cum-tunnel, reckoned that, above all, it would strengthen economic ties across the strait and create, within a few years, one of the fastest-growing and richest

regions in Europe. But ministers on both sides of the water, especially in Sweden, have been getting edgy about the bridge's teething problems. Last month Leif Pagrotsky, Sweden's trade minister, called for a tariff review: the cost of driving over the bridge, at SKr255 (\$26.40) each way, was too high to help integrate the region's two bits.

Businessmen have been complaining too. Novo Nordisk, a Danish drug firm which moved its Scandinavian marketing activities to Malmo to take advantage of 'the bridge effect', has been urging Danish staff to limit their trips to Malmo by working more from home. Ikea, a Swedish furniture chain with headquarters in Denmark, has banned its employees from using the bridge altogether when travelling on company business, and has told them to make their crossings – more cheaply if a lot more slowly – by ferry.

The people managing the bridge consortium say they always expected a dip in car traffic from a

summer peak of 20,000 vehicles a day. But they admit that the current daily flow of 6,000 vehicles or so must increase if the bridge is to pay its way in the long run. So they are about to launch a new advertising campaign. And they are still upbeat about the overall trend: commercial traffic is indeed going up. The trains have carried more than 1 m passengers since the service began in July.

Certainly, the bridge is having some effect. Many more Swedes are visiting the art galleries and cafés of Copenhagen; more Danes are nipping northwards over the strait. Some 75% more people crossed the strait in the first two months after the bridge's opening than during the same period a year before.

Other links are being forged too. Malmö's *Sydsvenska Dagbladet* and Copenhagen's *Berlingske Tidende* newspapers now produce a joint Oresund supplement every day, while cross-border

ventures in health, education and information technology have begun to bear fruit. Joint cultural ventures are also under way.

And how about linking eastern Denmark more directly with Germany's Baltic Sea coastline, enabling Danes to go by train from their capital to Berlin in, say, three hours? Despite the Danes' *nej* to the euro, it is still a fair bet that this last much-talked-about project will, within ten years or so, be undertaken.

Questions

- 1 Explain why the demand for the bridge is likely to be price-elastic.
- 2 If the Swedish government estimates that the price elasticity is -1.4 , calculate the effect on traffic using the bridge, stating any assumptions.
- 3 Why is the calculation above not likely to give an accurate forecast for the long term?

Case study 3.3: The Texas state bird

Southwest Airlines is a major carrier based in Texas, and has made a strategy of cutting fares drastically on certain routes with large effects on air traffic in those markets. For example on the Burbank–Oakland route the entry of Southwest into the market caused average fares to fall by 48 per cent and increased market revenue from \$21,327,008 to \$47,064,782 annually. On the Kansas City–St Louis route, however, the average fare cut in the market when Southwest entered was 70 per cent and market revenue fell from an annual \$66,201,553 to \$33,101,514.

Questions

- 1 Calculate the PEDs for the Burbank–Oakland and Kansas City–St Louis routes.
- 2 Explain why the above market elasticities might not apply specifically to Southwest.
- 3 If Southwest does experience a highly elastic demand on the Burbank–Oakland route, what is the profit implication of this?
- 4 Explain why the fare reduction on the Kansas City–St Louis route may still be a profitable strategy for Southwest.

Case study 3.4: Oil production

OPEC's oil shock³²

OPEC has surprised the markets with an output cut of 900,000 barrels per day, to take effect at the beginning of November. Observers had expected the oil producers' cartel to hold its quotas steady because production in Iraq has been hit by sabotage.

Before the regular meeting of the Organisation of Petroleum Exporting Countries (OPEC) in Vienna on Wednesday September 24th, most of the drama was

provided by Hugo Chávez, the Venezuelan president, who opined that the Iraq representative should not have been at the get-together because he was an illegitimate stooge of American occupiers. If that is so, Ibrahim Bahr al-Uloum behaved very oddly. His bullish predictions that Iraq could produce at least 3.5m barrels per day (bpd) by 2005 seem to have been among the factors that persuaded the ten members of OPEC's quota system to approve a surprise production cut of 900,000 bpd, to 24.5m bpd.

The effect of the cut was to send oil prices sharply higher. Equities in America retreated on fears that a higher oil price could stymie the incipient economic recovery: the Dow Jones Industrial Average of 30 leading shares fell by 1.57% that day.

In their official communiqué, OPEC's oil ministers pointed to their expectation of a 'contra-seasonal stock build-up' at the end of this year and the beginning of next year. Normally, oil stocks decline over the winter in the northern hemisphere, thanks to heavy use of heating oil. But this year, demand for oil, according to OPEC, will grow merely at its 'normal, seasonal' level, despite an improving world economy. OPEC expects supply to grow faster than demand, thanks to continued increases in production from Iraq and non-OPEC countries (of which Russia, the world's second-biggest oil exporter, is the most important).

OPEC expects this supply-demand mismatch to translate into a stock increase of 600,000 bpd in the final quarter of this year. This contrasts with an estimated stock reduction of 500,000 bpd in the final quarter of 2001, and 1m bpd in the last quarter of 2002. Larry Goldstein, president of the Petroleum Industry Research Foundation, believes OPEC has got its sums wrong. In remarks to the *Wall Street Journal*, he said he thought stocks would be flat over the coming three months.

Although the communiqué did not explicitly say so, OPEC members are keen to keep worldwide oil stocks below their ten-year average. That would give the cartel more power to determine the price. American oil stocks have been creeping up again after hitting 26-year lows earlier this year. America's energy secretary, Spencer Abraham, was clearly disappointed by OPEC's move, saying: 'Sustained global economic growth requires abundant supplies of energy. The US believes oil prices should be set by market forces in order to ensure adequate supplies.' America's opposition Democrats have been even more outspoken. Last month, they publicly rebuked Saudi Arabia, OPEC's (and the world's) leading producer, for reducing exports in August, thus causing an unpopular rise in American petrol prices.

Some observers are also speculating that OPEC may be sneakily trying to shift its price target above the current \$22–28 range (per barrel, for a basket of Middle Eastern crudes, which tend to trade a couple of dollars below West Texas crude). After all, the oil price has been well within that range for the past few

months. Why cut production when current supply levels are achieving their aim? In fact, the oil price has stayed higher than many expected: it was widely expected to fall well below \$20 per barrel after the end of the Iraq war. However, unrest in Nigeria, a big producer, and the continuing attacks on Iraq's oil facilities put paid to that.

OPEC's fears about non-OPEC production may be well-founded. After decades of communism, the industry in Russia is ramping up output: so far this year, it has been pumping an average of 800,000 bpd more than last year. Oil and gas are the country's biggest exports, earning hard currency that is seen as a key ingredient of economic revival. Moreover, the oil industry is in private hands, so even if the government in Moscow wanted to put a lid on production, it has less influence over its oil companies than OPEC governments have over theirs. The president of OPEC, Abdullah bin Hamad al-Attiyah, told the *Wall Street Journal* that the cartel would not cut production below 24m bpd unless big oil exporters outside OPEC, including Mexico and Norway as well as Russia, were prepared to cut production too.

OPEC's stance on Iraq is very different. Here, the cartel seems to be taking an overly rosy view. Iraq says it is currently producing around 1.8m bpd, well below the 2.5m bpd that it was pumping before the country was invaded in the spring (and even that was well below its potential, owing to years of sanctions). Moreover, exports, which are a crucial source of revenue for reconstruction, are still running at only about 500,000 bpd, compared with 2m bpd before the war. These have been seriously disrupted, and continue to be threatened by sabotage. Currently, oil is being exported mainly through the north: the southern ports on the Gulf coast are operating far below capacity.

For those who take OPEC's optimistic view of Iraqi production at face value, the cartel's move should not have come as a surprise. But the sharp reaction from oil markets and stockmarkets suggests it did. Many speculators had sold oil in the futures market, or 'shorted' it, expecting the price to fall in the short term – they clearly weren't expecting a big cut in output quotas any time soon. According to the Commodity Futures Trading Commission, the American regulator for commodity futures markets, the increase in short positions over September was equivalent to 470,000 barrels of oil. OPEC's decision led to a scramble to 'cover' such positions by buying oil. Whether prices

stay higher will depend on two key factors. Will OPEC members stick to their new quotas? (They have a history of cheating.) And will Iraqi militants continue to destroy their own country's wells and pipelines?

Questions

- 1 OPEC currently produces about 38 per cent of the world output of oil. Assuming the short-term price elasticity of demand is -0.28 , estimate the effect of the output cut on the current price, stating any assumptions in your calculations.
- 2 Describe the factors currently driving the world demand for oil; why has the price not fallen below the \$20 level as many expected?
- 3 Explain the effect of other non-OPEC producers on the cartel's output decisions.

Summary

- 1 The term 'demand' can be used in many contexts, as in demand schedules, curves, equations, functions and quantity demanded. It is important to distinguish between these different meanings.
- 2 Graphs and equations can be drawn and written with either price or quantity as the dependent variable.
- 3 Coefficients in demand equations can have either graphical or economic interpretations.
- 4 Marginal effects and elasticities are two different ways of describing the effect of a variable on quantity demanded.
- 5 Indifference curve analysis can be used to determine the inverse relationship between price and quantity demanded.
- 6 Price changes have two simultaneous effects on quantity demanded: an income effect and a substitution effect. The size and direction of these will determine the PED.
- 7 Although the basic neoclassical analytical framework and assumptions have certain limitations, the model is very versatile and is capable of being extended to apply to more complex situations.
- 8 There are a large number of factors which affect demand in reality; it is useful to distinguish between controllable and uncontrollable factors.
- 9 The concept of elasticity is vital in understanding and analysing demand relationships.
- 10 Theoretical considerations relating to the sign and size of elasticities must always be tested empirically.

Review questions

- 1 Explain the problems for government policy if it tries to use supply-oriented policies rather than demand-oriented ones in trying to discourage the consumption of certain products.

- 2 Explain the significance for economic theory of the relationship between price and perceived quality.
- 3 Analyse the effects on the demand for cars of the following:
 - a. A higher tax on gasoline
 - b. A tax on car parking
 - c. Increased automation on subways
 - d. Government legislation for increased safety in cars, like side impact protection
 - e. A 'congestion' tax imposed on cars using certain routes at certain times.
- 4 Select a company from the FT.com website and analyse the factors affecting its demand according to the categories described in this chapter.
- 5 Referring to Table 3.4 showing empirical estimates of price elasticities, draw some general conclusions relating to the products involved. Identify any apparent anomalies in the table and give some possible explanations.
- 6 Referring to Table 3.6 for income elasticities, repeat the procedure in the previous question.

Problems

- 3.1 BAD Enterprises is considering increasing the price of its harmonicas, currently \$20, by 25 per cent. BAD's current revenue is \$12,000 a month, and the PED for its harmonicas is estimated to be -1.8 .
 - a. Calculate the effect of the price change on BAD's revenue.
 - b. BAD now considers increasing its advertising budget to restore its sales revenue to its previous level. BAD is currently spending \$1,500 a month on advertising and estimates its AED to be 1.5. What will its new budget have to be?
 - c. What can you say about what will happen to profit in both (a) and (b) compared with the original level of profit?
- 3.2 Vik and Fleet produce trainers in the sports-shoe market. For one of their main products they have the following demand curves:

$$\text{Vik } P_V = 175 - 1.2Q_V$$

$$\text{Fleet } P_F = 125 - 0.8Q_F$$

where P is in £ and Q is in pairs per week.

The firms are currently selling 80 and 75 pairs of their products per week respectively.

- a. What are the current price elasticities for the products?
- b. Assume that Vik reduces its price and increases its sales to 90 pairs and that this also causes a fall in Fleet's sales to 70 pairs per week. What is the cross-elasticity between the two products?

- c. Is the above price reduction by Vik to be recommended? Explain your answer.
- 3.3 R and B Railroad decided to reduce their fares to see if it would help their business. They reduced their ticket prices by 25 per cent on average and their revenues increased from \$325,000 per month to \$332,000 on average.
- Without doing any calculations, comment on the PED.
 - Calculate the PED.
 - R and B decided that the fare reduction was a failure; why might they have done this?
- 3.4 Sales of cars declined by 15 per cent in a recent recession. During this period the price of cars rose by 6 per cent, average income fell by 4 per cent and the price of petrol rose by 20 per cent. It has been estimated that the PED for cars is - 0.8 and the YED is 1.25.
- Estimate the effect of the decline in income on car sales.
 - Estimate the effect of the car price rise on sales.
 - Estimate the CED between petrol and cars; compare your estimate using simple elasticities and additive effects with the estimate obtained using a power form of demand equation.
- 3.5 MK Corp estimates that its demand function is as follows:

$$Q = 400 - 12.5P + 25A + 14Y + 10P^*$$

where Q is the quantity demanded per month, P is the product's price (in \$), A is the firm's advertising expenditure (in \$'000 per month), Y is per capita disposable income (in \$'000), and P^* is the price of AJ Corp.

- During the next five years, per capita disposable income is expected to increase by \$5,000 and AJ is expected to increase its price by \$12. What effect will this have on the firm's sales volume?
- If MK wants to change its price by enough to offset the above effects, by how much must it do so?
- Compare the profitability of maintaining sales volume by either changing price or changing advertising spending.
- If MK's current price is \$60 and it spends \$10,000 per month on advertising, while per capita income is \$25,000 and AJ's price is \$70, calculate the price elasticity of demand with the price change.
- What can be said about the effect of the above price change on profit?
- What can be said about the relationship between the products of MK and AJ?

Notes

- 1 A. Marshall, *Principles of Economics*, London: Macmillan, 1890.
- 2 J. R. Hicks, 'The valuation of social income', *Economica*, n.s., 7 (1940): 105-124.